SNP and InDel specific filtering of variant calls

Raw variant calls from GATK HaplotypeCaller

Separate SNPs and InDels
because variant classes have different error profiles

SNP filtering thresholds

- QD ≥ 1.8
- FS ≤ 80
- SOR ≤ 4.0
- MQ ≥ 40

MQRankSum ≥ −15
ReadPosRankSum ≥ −10

Purpose: remove SNPs with poor quality-by-depth, strand bias, weak mapping support, or read-position bias

InDel filtering thresholds

- QD ≥ 1.8
- FS ≤ 200
- SOR ≤ 10
- ReadPosRankSum ≥ −20.0

More relaxed thresholds used
for INDEL-specific calling sensitivity

Rationale: InDel calls are more affected by local alignment and sequencing-context errors than SNPs

Filtered high-confidence SNP and InDel datasets
used for variant-effect prediction and downstream interpretation